Aniket Saha

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Internship

Cognizant Technology Solutions
Summer Intern Jun'26 – Aug'26

- Contributed to the development of AI- and automation-driven solutions for enterprise workflows within a client delivery team.
- Worked on multiple AI use cases involving **Large Language Models**, **prompt engineering**, and workflow automation.
- Collaborated with cross-functional team members to analyze requirements and **develop AI-powered** proof-of-concepts for enterprise applications.

Projects

AI Friend | Python, Rust, Ollama, NATS, LiveKit, Qdrant, Neo4j Oct'25 – Present

- Architected a **local-first, multi-agent cognitive system** with **8 specialized agents** coordinated via **NATS JetStream**, integrating **PAD affect, neuromodulation, cognitive appraisal, and utility-based decision-making** for adaptive LLM behavior.
- Engineered a **hybrid episodic-semantic memory architecture** using **PostgreSQL/pgvector, Qdrant, and Neo4j**, incorporating ACT-R-inspired retrieval and relevance ranking.
- Developed an **interruptible, real-time voice interaction pipeline** using **Rust, whisper.cpp, LiveKit/WebRTC, and GPT-SoVITS**, supporting voice cloning, speech-emotion perception, affect-driven prosody, and streaming audio with barge-in handling.

  Github Repository Link: https://github.com/PALabs-v1/AI_friend

SRA | Next.js 16, TypeScript, Prisma 7, PostgreSQL, Upstash, Terraform, Docker Sep'25 – Present

- Developed an **AI-powered requirements engineering platform** supporting **4 LLM providers** (Gemini, OpenAI, Claude, Grok) and **4 specification standards**, with **32K-token RAG context** and automated prompt compaction for efficient generation.
- Architected a **5-stage serverless pipeline** orchestrating **5 specialized AI agents** for requirements synthesis and validation, integrating **6Cs quality audits**, an **85/100 quality gate**, and checkpoint-based recovery with Upstash QStash.
- Engineered a **bidirectional Spec-to-Code CLI** with automated codebase reverse engineering and requirement traceability, integrating **pgvector hybrid search**, **2-level DFD generation**, self-healing UML diagrams, and version-controlled specifications.

  Github Repository Link: https://github.com/Aniket-a14/SRA

Wizard-w2 | Python, FastAPI, Next.js 16, Go, Ollama, Docker, SQLite-vec May'25 – Present

- Architected a **local-first autonomous data analysis agent** with **3 configurable model roles** (Manager, Worker, Embeddings), **6 intent-aware workflows**, and **3 data modes** (Local, Hybrid, Cloud), enabling adaptive planning, Python/SQL execution, and iterative verification.
- Engineered **2 isolated execution backends** (OS-sandboxed subprocess and Docker) with AST security enforcement, traceback-driven self-correction, and multi-agent **Council** reviews; validated **12/12 adversarial code-guard tests**.
- Developed a **cross-platform Go CLI** supporting **5 release targets** across macOS, Linux, and Windows, integrating real-time WebSocket streaming, **7 tabular data formats**, and hybrid **SQLite-vec/BM25 retrieval** for context-aware analysis.

  Github Repository Link: https://github.com/Wizard-AIA/Wizard-w2

Certifications

- **Social Networks | NPTEL** Nov'25
  Studied **graph theory**, **network analysis**, and **information diffusion models**, enabling deeper understanding of **information flow**, **system interactions**, and **data relationships** in large-scale systems.

Certificates

- **TCP/IP and Advanced Topics | Coursera** Nov'24
  Studied **TCP/IP protocols**, **network layers**, and advanced communication concepts, enabling deeper understanding of **distributed systems**, **network interactions**, and real-world system behavior.
- **Digital Systems: From Logic Gates to Processors | Coursera** Oct'24
  Learned **digital logic design**, **processor architecture**, and low-level computation, strengthening system-level reasoning and understanding of how software interacts with hardware.
- **Basic Python towards ML/AI | CSE Pathshala** Mar'24
  Completed a **25-hour live training program** covering Python fundamentals and introductory ML concepts, building practical understanding of **programming for data processing**, **problem solving**, and foundational AI workflows.

Achievements

- Active **open-source contributor**, publishing reusable components and contributing to open-source projects on **GitHub** alongside to the **Coder Legion** community, collaborating with practical experience in codebases. *Nov' 25 – Present*
- Solved **1100+** algorithmic problems across **LeetCode**, **HackerRank**, **CodeChef**, **Codeforces**, and **GeeksforGeeks**, strengthening **data structures**, **algorithms**, and consistent problem-solving ability across competitive programming platforms. *Dec' 24 – Present*
- Research paper titled **"A Novel Framework for Improved Cognitive, Emotional, and Vocal Interaction in Conversational Humanoid Robots Assisted by Hybrid Neuromodulatory Network Architecture"** published in **IEEE Xplore** as part of **ICoICI 2026** conference proceedings. **[View Publication]** *Oct' 26*

Techincal Skills

Languages: Python, TypeScript, JavaScript
Frameworks: React, Next.js, FastAPI, Express.js
AI/ML: LLM Agents, RAG Pipelines, Prompt Engineering, Ollama
Systems & Tools: Docker, Terraform, Git, Postman, Kubernetes, Linux
Databases: PostgreSQL, MySQL, MongoDB, Prisma ORM

Education

Lovely Professional University Phagwara, Punjab
Bachelor of Technology in Computer Science and Engineering - CGPA: 7.72 Aug'23 – Present

Indira Gandhi Memorial High School Barasat, West Bengal
Intermediate - Percentage: 75% Mar'22 – May'23

St. Stephen's School Habra, West Bengal
Matriculation - Percentage: 86% Mar'20 – May'21